

UTSAV BK

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EDUCATION

Howard University

Bachelor of Science in Computer Science

Washington, D.C.

Aug 2026 – Present

- **Honors & Fellowships:** Howard University Presidential Scholar (Full Merit Scholarship recipient).

Budhanilkantha School

Cambridge International A-Levels

Kathmandu, Nepal

Jun 2023 – Jun 2025

- **Academic Performance:** Physics (A*), Chemistry (A), Mathematics (A), Biology (A), General Paper (a).
- **Honors & Testing:** Morgan Award Winner 2025 (Highest A-Level Biology score cohort-wide); SAT: **1590/1600** (Math: 790, Reading/Writing: 800).

TECHNICAL PROJECTS & RESEARCH

SATitude (satitude.xyz)

Founder & Lead Full-Stack Developer

Jul 2026 – Present

Remote

- Architected and deployed a full-stack, AI-powered SAT prep platform with adaptive diagnostic tests and real-time performance analytics.
- Engineered an adaptive AI tutor utilizing OpenAI REST APIs, Node.js, Express, and PostgreSQL for real-time streaming feedback.
- Optimized client-side rendering and SQL query indexing, reducing page load latency by 35% and maximizing search visibility.

Hexapod Autonomous Robot

Hardware & Software Developer – Budhanilkantha Makerspace

Jun 2024 – Feb 2025

Kathmandu, Nepal

- Engineered and assembled a 6-legged autonomous robot under faculty mentorship, integrating microcontrollers and multi-sensor arrays.
- Programmed inverse kinematics and gait-coordination algorithms in Python for smooth multi-axis leg movements and sensor processing.
- Designed custom 3D structural components in Blender and resolved low-level hardware-software timing integration challenges.

Applied Mathematics Research Cohort

Mathematics Research Team Member – Guidelight Nepal

Nov 2024 – Apr 2025

Kathmandu, Nepal

- Selected for a selective 7-person research cohort guided by academic mentors from Cornell University and the University of Bristol.
- Investigated advanced functional transformations and non-linear parameter behavior using analytical mathematical modeling.

TEACHING & LEADERSHIP EXPERIENCE

Merosiksha

Online STEM Tutor

Aug 2025 – May 2026

Kathmandu, Nepal

- Delivered comprehensive 1-on-1 and group instruction in Physics, Chemistry, Math, and Biology on Nepal's premier online learning platform.
- Designed tailored practice problem sets and step-by-step conceptual breakdowns to improve student mastery in STEM curricula.

Environment Club, Budhanilkantha School

Vice President

Feb 2024 – Feb 2025

Kathmandu, Nepal

- Directed a year-long environmental initiative engaging 1,100+ students through interactive workshops with Kathmandu District Forest Office.

Navigo Nepal ("What's Next" Program)

Jan 2025 – May 2025

- Educated 500+ high school students across Kathmandu on post-secondary academic pathways, STEM olympiads, and global scholarships.

TECHNICAL SKILLS & SPOKEN LANGUAGES

Languages:	Python, C++, JavaScript (ES6+), SQL, HTML5, CSS3
Frameworks & Tools:	React, Node.js, Express, PostgreSQL, REST APIs, Git, GitHub, Blender (3D Modeling)
Core Concepts:	Data Structures & Algorithms, Kinematics & Robotics Control, Asynchronous Data Flow
Spoken Languages:	English (Fluent), Nepali (Native), Hindi (Fluent)